

Nicolas Kessler

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Education

Rochester Institute of Technology (RIT)

Bachelor of Science in Mechanical Engineering, Minor in Mathematics

GPA: 3.72 / 4.0 | Honors Program | RIT Presidential Scholar | Pi Tau Sigma

Rochester, NY

Expected May 2027

Relevant Coursework: Engineering Measurements Lab, Thermodynamics, Strength of Materials I & II, Fluid Mechanics, System Dynamics, Dynamics, Materials Science, Multidisciplinary Senior Design

Skills

Software: MATLAB, Simulink, SolidWorks, Siemens NX, ANSYS, LabVIEW, Python, C, Revit, AutoCAD, Excel

Technical: GD&T, Precision Metrology ($\pm 1 \mu\text{m}$), 3D Printing (SLA/FDM), Manual Milling, Soldering, Experimental Testing, Data Analysis, Mechatronics, Root Cause Analysis, Quality Control

Languages: English (Fluent), Spanish (Fluent), French (Fluent), Italian (Intermediate)

Experience

BNP Associates

Fairfield, CT (New York City Metro Area)

Project Engineer Intern

May 2026 – August 2026

- Produced **30 design-option layouts** in **Revit** across **8 airport infrastructure projects**, delivering alternatives directly to client teams to accelerate layout selection
- Cataloged **17 ground-support vehicles** for **AutoCAD Vehicle Tracking** simulations, improving the efficiency and accuracy of swept-path studies
- Recreated a client-provided AutoCAD model of an entire airport terminal in Revit, translating CAD geometry into a **BIM-ready model** used for downstream design coordination
- Ran **4 swept-path studies** using AutoCAD Vehicle Tracking to validate airport layouts against vehicle turning and clearance requirements, informing design approval decisions

Clarapath

Hawthorne, NY (New York City Metro Area)

Production Engineer Co-op

June 2025 – December 2025

- Performed quality control and documentation on **300+ precision components** using advanced metrology tools; ranked **#1 among co-ops** in throughput and speed
- Performed precision inspections to $\pm 1 \mu\text{m}$ tolerances and managed **50% of departmental QC tasks** while training **5 peers** on inspection protocols
- Designed and 3D printed standardized laser calibration blocks in **SolidWorks**, improving calibration consistency and testing flexibility
- Developed a shock-preventing equipment cover **adopted company-wide** and authored QC training manuals to standardize inspection and training procedures

Copeland

Sidney, OH

Applied Mechanics Co-op

August 2024 – December 2024

- Independently owned **9 mechanical test programs** end-to-end, from fixture design and instrumentation selection to **LabVIEW** data acquisition, **Python/MATLAB** analysis, and formal reporting
- Redesigned testing procedures to consolidate **5 impact hammers into 1**, resulting in a **5x cost reduction** and improved data accuracy through integrated temperature sensing
- Developed capacitor- and load cell-based measurement systems for clamp-load and distance testing; validated measurements against a **10-kip MTS load frame** and developed procedures adopted by the Reliability Team
- Performed torque-to-yield testing on DIN rail clamps using **LabVIEW** data acquisition and **Python** analysis to statistically compare new and existing designs

Projects

High-Velocity Vacuum Launcher

- Developed a physics-based model using **fluid dynamics and kinematics** to predict projectile exit velocity from pressure differential
- Co-developed a **MATLAB simulation** and experimentally validated predicted exit velocities using instrumented launcher testing

Automated Percussion Machine

- Designed a **mechatronic system** using **Arduino and C** to automate musical performances
- Calculated **acoustic resonant frequencies** of piping to determine precise lengths required for specific musical notes
- Integrated **stepper motors** with custom driver logic to control striking velocity and damping for consistent tonal quality